# AM5700: Sugar and Spice and Everything Nice

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**EP23B049**

The objective of this work is to develop an agent-based model to study inter-tribe dynamics arising from competition over limited resources. In particular, the model investigates the onset of conflict as a consequence of resource pressure, and develops a comprehensive set of combat strategies built upon the foundational framework of *Sugarscape* introduced by J. M. Epstein and R. L. Axtell in *Growing Artificial Societies: Social Science from the Bottom Up* (MIT Press, 1996). Economic indicators such as wealth distribution and inequality are employed to quantify dominance between competing groups.

## 1 The Set-Up

- **Grid:** $50 \times 50$ lattice with non-periodic boundaries.
- **Resources:** Sugar and spice distributed as spatial "mountains" (inspired by Epstein and Axtell (1996)), with sugar peaks at $(15, 15)$, $(35, 35)$ and a spice peak at $(15, 35)$; capacities decay radially and regenerate each timestep.
- **Agents:** Instances of a TribeAgent class with attributes: position, vision, metabolism, resource endowments, age, and a binary tag defining tribal identity.

## 2 Agent Interactions (Peacetime)

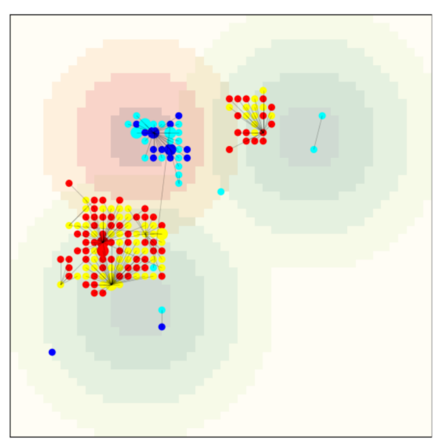
In the absence of conflict, agent dynamics are governed by local movement, trade, reproduction, and taxation.

- **Movement:** Agents move within their vision to maximize
$$U = \sqrt{(s+1)(p+1)},$$
with social Schelling-like weighing
$$W = U \cdot \frac{1 + \text{allies}}{(1 + \text{enemies})^2}.$$
- **Trade:** MRS given by
$$\text{MRS} = \frac{p}{s}, \quad \pi = \sqrt{\text{MRS}_1 \cdot \text{MRS}_2},$$
with trades executed only if welfare increases for both agents.
- **Reproduction:** Occurs locally under resource and age constraints, with inherited traits.
- **Taxation:** $\tau = rs$, with policy 0 (none), 1 (to leaders), and 2 (redistribution).

## 3 Wartime Dynamics

Conflict is triggered when the total population exceeds a threshold $N > 600$, at which point agents switch from economic optimization to competitive interactions.

- Target Selection: Agents evaluate enemies within their vision using a resource-weighted score
$$Q = \alpha s + (1 - \alpha)p,$$
where $\alpha$ encodes tribal preference. Agents preferentially engage targets with higher $Q$.
- **Combat Strength:** Each agent is assigned a strength
$$S = (\max(v - m, 0) + 1) \cdot \min\left(\frac{s}{s_0}, 1\right),$$
where $v$ is vision, $m$ is metabolism, and $s_0$ is a reference resource level.
- **Group Scaling:** Strength is scaled by local group size, inspired by Lanchester scaling
$$S_{\text{eff}} = S \cdot \frac{n_{\text{own}}}{n_{\text{enemy}}},$$
capturing linear reinforcement effects in coordinated combat.
- **Combat Outcome:** Interactions are probabilistic, with agent $A$ winning against $B$ with probability
$$P(A \text{ wins}) = \frac{S_A}{S_A + S_B}.$$
The losing agent is removed, and a fraction of its resources is transferred to the winner.



*Emergence of Tribal Subcultures*

## 4 Economic Indicators

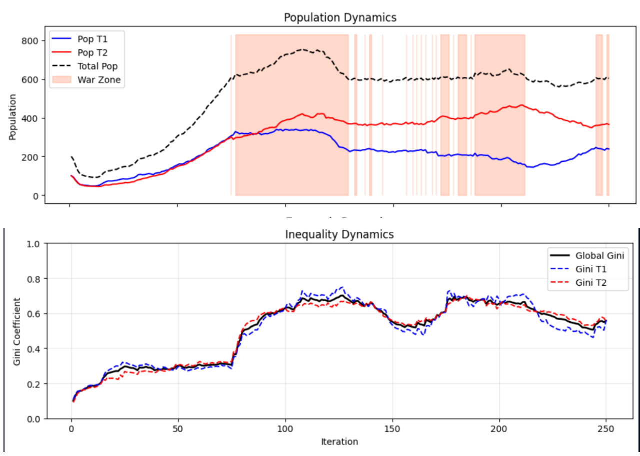
To quantify wealth inequality, we employ the Gini coefficient. For a population of $n$ agents with sorted wealth values $w_i$, it is computed as

$$G = \frac{2\sum_{i=1}^{n} i\, w_i}{n \sum_i w_i} - \frac{n+1}{n}.$$

The coefficient lies in $[0, 1]$, where $G = 0$ corresponds to perfect equality and larger values indicate increasing concentration of wealth. In this model, wealth is defined as $w_i = s_i + p_i$, capturing total resource holdings of each agent.

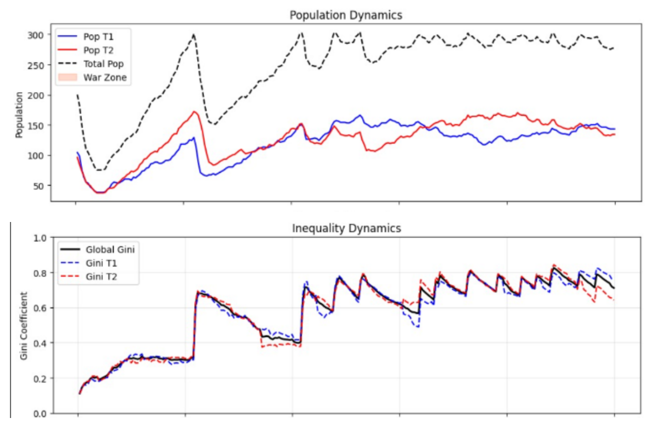
## 5 Emergent Dynamics

- **Early regime:** Both tribes exhibit population growth under peacetime reproduction. The Gini coefficient increases only marginally, as local trade (via MRS-based exchange) redistributes resources and suppresses inequality at the neighborhood scale.
- **Onset of conflict:** Once $N > 600$ and war is triggered, a sharp increase in the Gini coefficient is observed. Combat introduces winner-takes-most dynamics: successful agents accumulate resources from defeated opponents, while losers are eliminated. Since victory probability increases with effective strength (itself positively correlated with resource holdings), this creates a positive feedback loop in which already wealthy agents are more likely to win and become even wealthier, leading to rapid concentration of wealth and a spike in inequality.

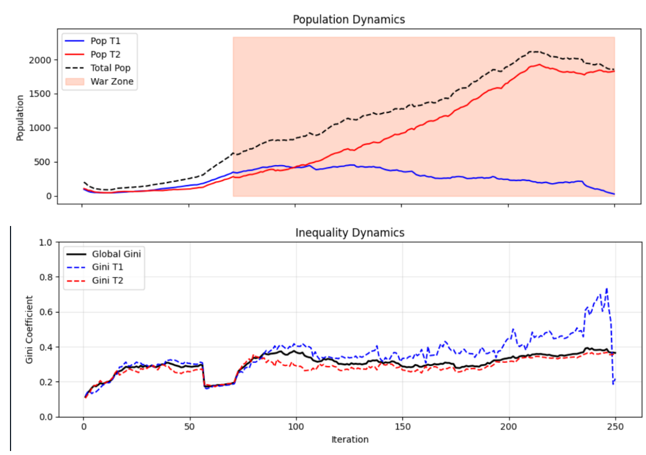



*Spiking in Gini inequality observed when war is triggered (pink shaded regions).*

- **Feudal regime** (tax policy 1 for both tribes): Leader-based taxation concentrates resources at the top, producing sustained high inequality. Population repeatedly crosses the war threshold, leading to cyclic war–peace dynamics. Correspondingly, the Gini coefficient exhibits sharp spikes during conflict and partial relaxation during peacetime.
- **Redistributive regime**(tax policy 2 for both tribes): Uniform redistribution suppresses large wealth disparities, resulting in comparatively lower and smoother Gini values. A noticeable dip in Gini is observed prior to war, reflecting increased redistribution as population grows and total resources expand. Conflict dynamics are less abrupt, and oscillatory behavior is significantly reduced.




*Feudal regime: Leader-based taxation concentrates wealth, resulting in sharp sparks in Gini inequality*




*Redistributive regime : Uniform redistribution amongst tribes reduces inequality.*

- Long time simulations show that unless a very careful set of parameters are chosen one tribe almost always dominates over the other, filling up the entire grid. This is visible in the last graph as well.

## 6 Contributions

- **Trade modelling:** Developed a complete agent-level trade mechanism grounded in marginal rates of substitution (MRS), defined as

$$\mathrm{MRS} = \frac{p}{s},$$

with trades executed at geometric mean price $\pi = \sqrt{\mathrm{MRS}_1 \mathrm{MRS}_2}$ under mutual welfare improvement. The model extends the original Epstein–Axtell framework to a two-resource (sugar–spice) economy with local interactions and endogenous price formation.

- **Economic characterization:** Introduced the Gini coefficient,

$$G = \frac{2\sum_{i=1}^{n} i\, w_i}{n\sum_{i=1}^{n} w_i} - \frac{n+1}{n},$$

as a global indicator of inequality, with $w_i = s_i + p_i$. Used Gini dynamics to systematically characterize the impact of taxation regimes and conflict on wealth distribution, identifying distinct signatures such as pre-conflict equilibration and wartime concentration.

- **Combat modelling:** Formulated a combat framework combining intrinsic agent attributes and resource dependence. Agent strength is modeled as

$$S = (\max(v - m, 0) + 1) \cdot f(s),$$

where $v$ is vision, $m$ is metabolism, and $f(s)$ encodes a resource-dependent health multiplier. Group interactions incorporate Lanchester-type scaling,

$$S_{\text{eff}} \sim S \cdot \frac{n_{\text{own}}}{n_{\text{enemy}}},$$

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capturing the effect of coordinated group size on combat outcomes.

- **Simulation and validation:** Implemented the full agent-based model in Python, integrating trade, taxation, reproduction, and combat dynamics. Performed most of primary coding work. Performed extensive validation to ensure consistency of spatial interactions (Moore neighborhoods), vision constraints, distance metrics, and state updates across all components. Multiple simulation runs were conducted to verify robustness of emergent behaviors under varying parameter regimes.

## 7 Code

The complete implementation of the model, including simulation and visualization scripts, is available at:

- Project Code Repository

## 8 References


- J. M. Epstein and R. L. Axtell, *Growing Artificial Societies: Social Science from the Bottom Up*, MIT Press (1996).
  Link to book
- J. M. Epstein, *Modeling Civil Violence: An Agent-Based Computational Approach*, *Proceedings of the National Academy of Sciences*, 99 (suppl_3), 7243–7250 (2002).
  Link to paper
- Lanchester-type scaling in combat models.
  Lecture notes
- Gini coefficient (definition and interpretation).
  Wikipedia article